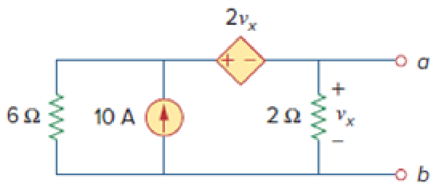
# Chapter 4, Problem 12PP

## Problem

Find the Norton equivalent circuit of the circuit in Fig. 4.45 at terminals *a*-*b*.

**Figure. 4.45:**




## Step-by-step solution

**Step 1** of 7

Refer to figure 4.45 given in the textbook.

Short circuit the terminals *a* and *b*. And the current flowing through the same short circuited path is $i_{sc} = I_N$ as shown in figure 1.

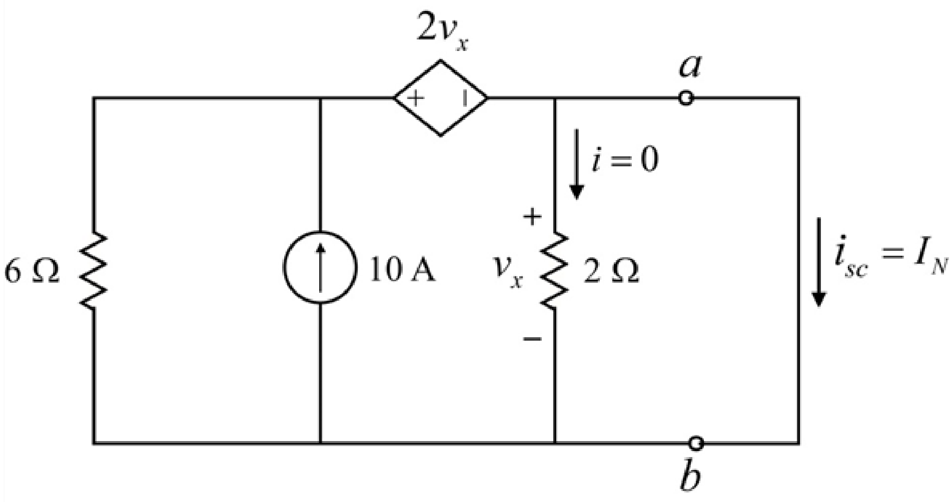



Figure 1

**Step 2** of 7

Consider the circuit shown in figure 1.

As no current flows through the resistance of $2\ \Omega$, open circuit it as shown in figure 2.

Also,

$v_x = 0$

Short circuit the dependent voltage source of $2v_x$ as shown in figure 2.

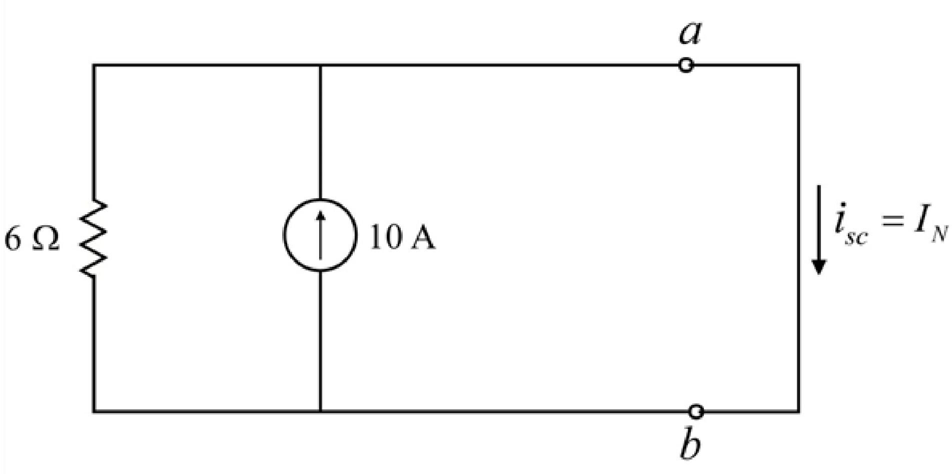



Figure 2

**Step 3** of 7

Consider the circuit shown in figure 2.

The current always flow in a low resistance path.

Therefore, the Norton current $I_N$ is $\boxed{10\text{ A}}$.

**Step 4** of 7

Refer to figure 4.45 given in the textbook.

Determine the Norton resistance $R_N$.

Place a voltage source of 1 V across the terminals *a* and *b* as shown in figure 3.

Open circuit the independent current source of 10 A as shown in figure 3.

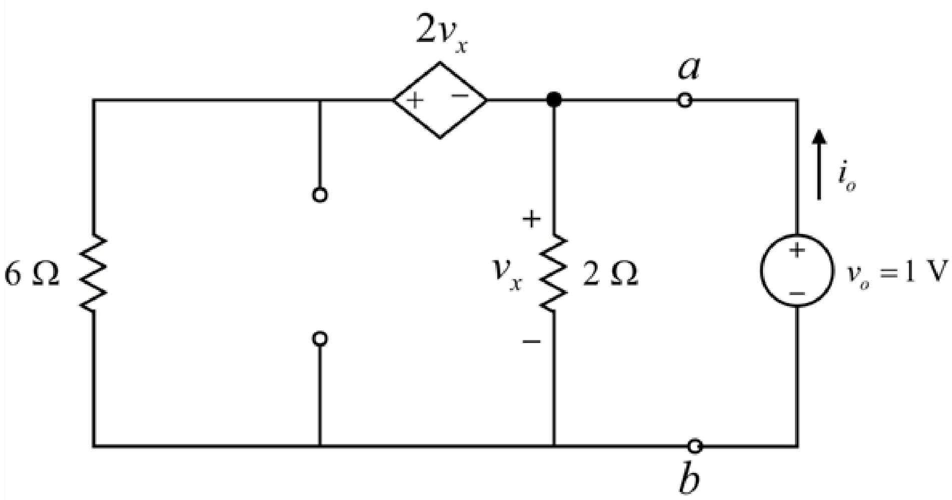



Figure 3

**Step 5** of 7

Apply Nodal analysis at node-*a*.

$$\frac{v_x + 2v_x}{6} + \frac{v_x}{2} - i_o = 0$$

Substitute 1 for $v_x$.

$$\frac{1+2(1)}{6} + \frac{1}{2} - i_o = 0$$

$$i_o = \frac{3}{6} + \frac{1}{2}$$

$$i_o = 1\ \text{A}$$

**Step 6** of 7

Calculate the Norton resistance $R_N$.

$$R_N = \frac{v_o}{i_o}$$

Substitute 1 for $i_o$, and 1 for $v_o$.

$$R_N = \frac{1}{1} = 1\ \Omega$$

Therefore, the Norton resistance $R_N$ is $\boxed{1\ \Omega}$.

**Step 7** of 7

Draw the Norton equivalent circuit as shown in figure 4.

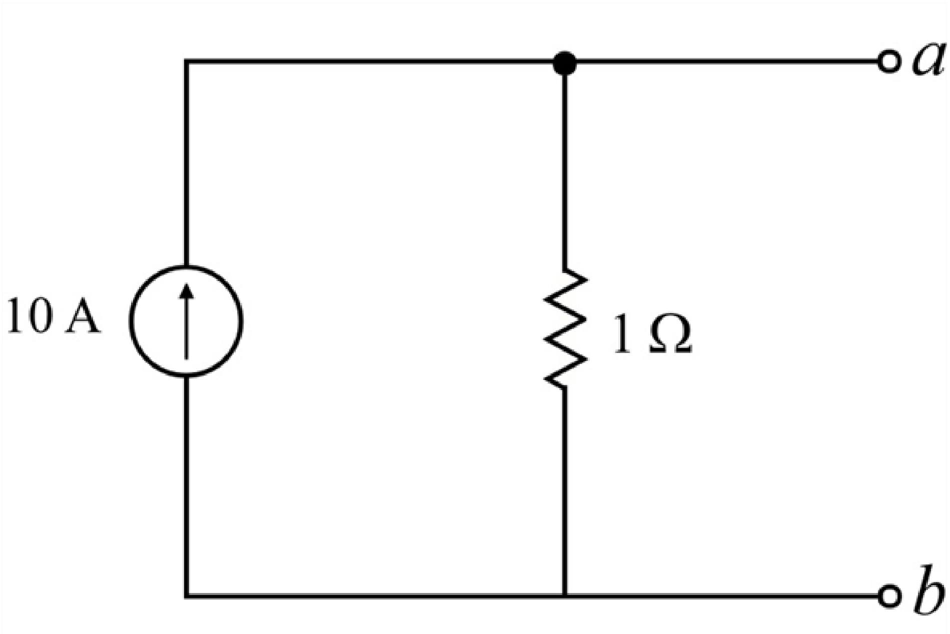



Figure 4